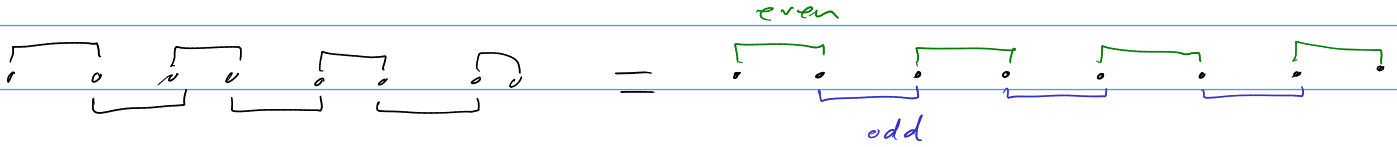


$$h = -it \sum_i \gamma_i \gamma_{i+1} + g \sum_i \gamma_i \gamma_{i+2} \gamma_{i+3} \gamma_{i+4} + \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3}$$

split hamiltonian into h_{even} , h_{odd} and rewrite majorana operators into even and odd:

$$a_j := \gamma_{2j}$$

$$b_j := \gamma_{2j+1}$$



$$h = h_{\text{even}} + h_{\text{odd}}$$

$$= -it \left(\sum_j a_j b_j + \sum_j b_j a_{j+1} \right)$$

→ Use Jordan Wigner transformations to map fermionic (majorana) operators to spin operators (Pauli matrices)

$$-it \left(\sum_j i \sigma_j^z + K_j \sigma_j^y K_{j+1} \sigma_{j+1}^x \right)$$

$$= -it \left(\sum_j (i \sigma_j^z + K_j \sigma_j^y K_j \sigma_j^z \sigma_{j+1}^x) \right)$$

$$-it \left(\sum_j i \sigma_j^z + \cancel{K_j} \sigma_j^y \sigma_j^z \sigma_{j+1}^x \right)$$

$$= i \sigma_j^x$$

$$\sigma_i \sigma_j = i \epsilon_{ijk} \sigma_k$$

$$= -it \left(\sum_j i \sigma_j^z + i \sigma_j^x \sigma_{j+1}^x \right)$$

$$= \sum_j -\cancel{t} \sigma_j^z - \cancel{t} \sigma_j^x \sigma_{j+1}^x = \sum_j -J \sigma_j^x \sigma_{j+1}^x - h \sigma_j^z \quad \text{as expected.}$$

$$1 \times t \quad t = J = h$$

$$a_j = K_j \sigma_j^x$$

$$b_j = K_j \sigma_j^y$$

$$a_j b_j = i \sigma_j^z$$

$$K_j = \prod_{l=1}^{j-1} \sigma_l^z$$

$$g \sum_i r_i r_{i+2} r_{i+3} r_{i+n} = g \sum_j \underbrace{a_j a_{j+1}}_{\text{blue}} \underbrace{b_{j+1} a_{j+2}}_{\text{blue}} + \underbrace{b_j b_{j+1} a_{j+2} b_{j+2}}_{\text{red}}$$

$$\begin{array}{|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet \\ \hline \end{array} = \begin{array}{|c|c|c|c|c|} \hline a_j & b_j & a_{j+1} & a_{j+1} & a_{j+2} & b_{j+2} \\ \hline \end{array}$$

$$a_j a_{j+1} b_{j+1} a_{j+2} =$$

$$K_j \sigma_j^x a_j \sigma_j^z - K_{j+2} \sigma_{j+2}^x = \cancel{K_j \sigma_j^x} a_j \sigma_{j+1}^z \cdot \cancel{K_{j+2} \sigma_{j+2}^x} \cdot \cancel{\sigma_j^z} \sigma_{j+1}^z \sigma_{j+2}^x$$

$$= \sigma_j^x a_j \sigma_{j+1}^z \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$a_j \sigma_j^x \sigma_j^z \sigma_{j+1}^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$i(-i \sigma_j^y) \sigma_{j+2}^x = \sigma_j^y \sigma_{j+2}^x$$

$$b_j b_{j+1} a_{j+2} b_{j+2} = \cancel{K_j \sigma_j^y} \cdot \cancel{K_{j+2} \sigma_{j+2}^z} \sigma_{j+1}^y \cdot i \sigma_{j+2}^z$$

$$\sigma_j^y \sigma_j^z \sigma_{j+1}^y i \sigma_{j+2}^z$$

$$\cancel{i \sigma_j^x} \sigma_{j+1}^y \sigma_{j+2}^z = - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z$$

$$g \sum r_i r_{i+2} r_{i+3} r_{i+n} = g \sum \sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z$$

$$+ \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3} = \sum_j \underbrace{a_j b_j a_{j+1} a_{j+2}} + \underbrace{b_j a_{j+1} b_{j+1} b_{j+2}}$$

$$\begin{array}{|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet \\ \hline \end{array} = \begin{array}{|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet \\ \hline \end{array}$$

$$\begin{aligned} \frac{a_j b_j}{\cancel{+}} \cdot a_{j+1} a_{j+2} &= i \sigma_j^z \cdot \cancel{h_j \sigma_j^z} \sigma_{j+1}^x \cdot \cancel{h_j \sigma_j^z} \sigma_{j+1}^z \sigma_{j+2}^x \\ &= i \sigma_j^z \cancel{\sigma_j^z} \sigma_{j+1}^x \sigma_{j+1}^z \sigma_{j+2}^x \quad \begin{array}{l} \gamma_j^z \\ \gamma_{j+1}^z \end{array} i \sigma_{j+1}^x \\ &= i \sigma_j^z (-i \sigma_{j+1}^y) \sigma_{j+2}^x = \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x \end{aligned}$$

$$\begin{aligned} b_j a_{j+1} b_{j+1} b_{j+2} &= \cancel{h_j \sigma_j^y} \cdot i \sigma_{j+1}^z \cdot \cancel{h_j \sigma_j^z} \sigma_{j+1}^z \sigma_{j+2}^y \\ &= \sigma_j^y \sigma_j^z (i \cancel{\sigma_{j+1}^z} \sigma_{j+1}^z) \sigma_{j+2}^y \\ &= (i \sigma_j^x)(i)(\sigma_{j+2}^y) = -\sigma_j^x \sigma_{j+2}^y \end{aligned}$$

$$g \sum \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3} = g \sum \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y$$

→ flipped sign of second term and reversed x, y, z index order
↳ makes sense.

$$h = -t \sum_j (\sigma_j^z + \sigma_j^x \sigma_{j+1}^x) + g \left(\sum_j (\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z) + (\sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y) \right)$$

$$h = -t \sum_j (\sigma_j^z + \sigma_j^x \sigma_{j+1}^x) + g \left(\sum_j (\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y) - (\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x) \right)$$

to study ferrimagnetic phases:

$$h = \sum_j J \sigma_j^x \sigma_{j+1}^x + h \sigma_j^z - g (\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y) - g (\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x)$$

MPO: Check first term:

$$h = \sum_j \sum_i c_j^x c_{j+1}^x + c_j^z$$

$$\begin{array}{cc|cc} 1 & 0 & \rightarrow & 0^7 & 0 \\ 0 & 1 & \rightarrow & 0^x & 0^x \\ -x-2 & -7 & & 1 & 0 \end{array}$$

2			x
1			x
0	x	x	x

Ans

$-x$	I	0	0
6^x	0	0	
I	6^x	6^x	1
	I	6^x	$-x$

1	1	1	✓
2	2	2	✓
3	3	3	✓
4	4	4	✓
5	5	5	✓
6	6	6	✓
7	7	7	✓
8	8	8	✓
9	9	9	✓
10	10	10	✓

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 6 \\ 1 & 6 & 6 \end{pmatrix}$

$$g(b_j^u, b_{j+2}^x - b_j^x, b_{j+2}^y)$$

The first diagram shows a 3D coordinate system with three axes labeled x , y , and z . The z -axis is vertical, the x -axis is horizontal, and the y -axis is diagonal. A curved arrow indicates a counter-clockwise rotation around the z -axis. The second diagram shows a 2D coordinate system with a horizontal x -axis and a vertical y -axis. A curved arrow indicates a counter-clockwise rotation around the origin.

$$\begin{array}{c|c|c} G^Y & G^X & I \\ \hline G^Y & I & G^X \\ \hline I & G^Y & I G^X \end{array} \quad \begin{array}{c|c|c} -G^X I & G^Y & I \\ \hline -G^X & I & G^Y \\ \hline I & -G^X & I G^Y \end{array}$$

I	I	-x-
-x-	I	I

Handwritten matrix reduction steps for a 7x7 matrix. The matrix is transformed through several steps, with rows and columns labeled with $\sigma^k I$.

Initial matrix (Step 1):

$$\begin{matrix} & 1 & 0 & 0 & 0 & 0 & 0 \\ \sigma^4 I & -6^0 & 0 & 0 & 0 & 0 & 0 \\ \sigma^2 I & 6^4 & 0 & 0 & 0 & 0 & 0 \\ \sigma^4 & 0 & -1 & 0 & 0 & 0 & 0 \\ \sigma^0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6^4 & 6^0 & 1 \end{matrix}$$

Intermediate matrix (Step 2):

$$\begin{matrix} & 1 & 0 & 0 & 0 & 0 & 0 \\ \sigma^4 I & -6^0 & 0 & 0 & 0 & 0 & 0 \\ \sigma^2 I & 6^4 & 0 & 0 & 0 & 0 & 0 \\ \sigma^4 & 0 & -1 & 0 & 0 & 0 & 0 \\ \sigma^0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6^4 & 6^0 & 1 \end{matrix}$$

Final matrix (Step 3):

$$\begin{matrix} & 1 & 0 & 0 & 0 & 0 & 0 \\ \sigma^4 I & -6^0 & 0 & 0 & 0 & 0 & 0 \\ \sigma^2 I & 6^4 & 0 & 0 & 0 & 0 & 0 \\ \sigma^4 & 0 & -1 & 0 & 0 & 0 & 0 \\ \sigma^0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6^4 & 6^0 & 1 \end{matrix}$$

The final matrix is in row echelon form, with a zero row at the bottom.

$$g(\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x)$$

$$\begin{array}{c|c|c} 1 & 1 & -x- \\ \hline 1 & G^0 & G^4 G^7 \\ \hline G^4 & G^4 & G^7 \\ \hline G^0 G^4 & G^7 & 1 \\ \hline -x- & 1 & 1 \end{array}$$

$$\begin{array}{c|c|c} I & -6^2 & 6^4 6^2 \\ 6^2 & 6^4 & 6^2 \\ 6^2 6^4 & 6^2 & I \end{array}$$

[illegible]

Combine all matrices together:

